

Clam It!

Game Design Document

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GitHub repository: <https://github.com/daggerscythe/clam-it>

Engine: Godot 4.7.2

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1. Game Concept

1.1 Elevator Pitch

Clam It! is a game where you grow, harvest, and sell pearls, while defending them from crabs. As you progress, you earn upgrades and new pearl types.

1.2 Concept Description

In the game, you plant dummy pearls and while waiting for them to mature, you have to keep them clean from sediment and safe from destruction by crabs. You can also use chum to boost the clam's growth, but the more you use it, the more crabs you attract. You then can sell harvested pearls for various amounts of money, depending on their rarity.

1.3 Connection to the Theme

The game is set entirely underwater, with access to a dock, where you sell your pearls.

1.4 Target Audience & Player Experience Goals

The game is a relaxing, resource & environment management simulator, targeted at players that enjoy similar upgrade-progression management games, like *Cookie Clicker*.

2. Genre

Primary Genres: farming simulator, casual management

3. Background Information

3.1 Story & Setting

You are a pearl farmer, trying to make some money selling pearls. Evil local crabs are trying to steal your pearls because they don't want to see you succeed (and they also like pearl-growing clams as food in real life).

3.2 Real-World Background

The core game features have been inspired by how things are in the real world: clams can grow pearls, crabs usually like to destroy clams, and all pearl types and rarities in the game were drawn from the real-world types.

3.3 What the Player Needs to Know Before Playing

- To launch the game, download the zip for your platform from the GitHub release and unzip it:
 - On Windows: double-click *ClamIt.exe*.
 - On macOS: right-click *ClamIt.app* and choose Open, then click Open again in the popup.
 - On Linux: make *ClamIt.x86_64* executable first, then double-click it or run `./ClamIt.x86_64`.
- There is only one usable save slot in the game, which is accessible through “save” and “load” in the pause menu:

- On Windows:
C:\Users\[USER]\AppData\Roaming\Godot\app_userdata\ClamIt\savegame.json
- On macOS: ~\Library\Application Support\Godot\app_userdata\ClamIt\savegame.json
- On Linux: ~\.local\share\godot\app_userdata\ClamIt\savegame.json
- If you want **to skip ahead and see everything**, there's a save with all the pearls unlocked in the repo at all_unlocked_save/savegame.json. It's at Level 16 with all 7 pearl types and all 6 clams unlocked, plus some money and sonic flares to play with. Copy it into the save folder for your platform (listed above, replace the old savegame.json if there is one), launch the game, then hit Esc and click Load.
- In the settings, there are three toggles:
 - Infinite sonic flares – you never run out of sonic flares.
 - Instant clean – using the brush instantly cleans the clam.
 - Instant chum – using chum on clam instantly matures the pearl.

4. What the Game Teaches

4.1 Learning Goals

Clam It! is mostly a game about juggling a bunch of small jobs at once and making smart calls about risk. Here's what I want players to walk away knowing:

- **Risk vs. reward:** chum makes pearls grow way faster, but every click fills up the chum meter, and once it passes the line, crabs show up. You have to figure out how much chum you can get away with.
- **Prioritizing:** once you have a few clams going, you're constantly deciding which one to scrub, which one to chum, and which one the crabs are heading for and where to use your limited sonic flares.
- **Timing and limited resources:** flares are limited (or cost money), so you learn to wait until crabs are close or bunched up before firing, instead of spamming them.
- **Spending money wisely:** in freeplay you have to decide between buying an upgrade now, saving it for later, or keeping cash for flares, and which upgrades actually fit how you play.
- **A little bit of real pearl farming:** pearls grow around a planted nucleus, clams need to stay clean to grow well, and pearl types really do differ a lot in rarity and value.

4.2 How Each Concept Is Taught

Concept / Ability	Where it is introduced	How the player is taught (popup, log, forced first use)
Planting & harvesting (Hand)	Level 1	A welcome popup shows up as soon as the game starts and explains the basics: planting, waiting for the pearl to grow, harvesting, and selling at the dock. You also start with the Hand already selected and only one clam, and growth only takes 2 seconds at Level 1, so you get to try it all almost right away. The chat log backs it up ("Planted pearl dummy!", "Pearl is still growing!") and the green bar shows progress.
Selling at the surface dock	Level 1	Also covered by the welcome popup. On top of that, the dock at the top of the screen has a [SELL HERE] sign, and clicking it with the Hand sells everything you're carrying. Floating text shows the money you made (or "No pearls to sell!"). Selling gives way more XP than anything else, so it's the thing that actually levels you up.
Scrubbing sediment (Brush)	Level 2	Level-up popup that pauses the game and shows the brush icon, how to use it (press 2 or B, click a dirty clam) and what it does. Sediment only starts building up at Level 2, and a second clam unlocks, so you need to scrub pretty much right away.
Chum speed boost & chum meter	Level 3	Level-up popup again. At Level 3 a third clam unlocks, so now you're waiting on three 10 second pearls at once and chum is the obvious way to speed things up. The chum meter starts filling in the top right, but nothing bad happens yet because crabs aren't unlocked.
Crabs & Sonic Flares	Level 4	Level-up popup plus 5 free sonic flares. Growth also goes up to 30 seconds here, so the chum you just learned to love gets even more tempting, and it now attracts crabs once the meter passes the white tick. The chat log warns you when a crab attacks and the clam's bar turns red, so it's clear what's going on.
Upgrades, shop, pearl types	Level 5+ / Freeplay	At Level 5 you get a popup for the new Blush pearl, then an upgrade pick popup where you choose a card and hit Buy Now or Save for Later. The shop button shows up at Level 4 (sonic flares in the Supplies tab) and the Available and Owned tabs fill in from Level 5.

5. Core Mechanics & How to Play

5.1 Controls

Input	Action
1 / H	Hand: plant dummy / harvest
2 / B	Brush: scrub sediment
3 / C	Chum: boost growth
4 / F	Sonic flare: scare crabs
Mouse click	Interact with UI / use the selected tool
Esc / Pause button	Pause the game & access the pause menu

5.2 Core Mechanics

Planting a pearl dummy

Select the Hand (1 or H) and click an empty clam (closed, no bar above it) to drop a pearl dummy in. That starts the growth timer and gives you 2 XP. Every pearl starts here, so this is step one of the whole game.

Pearl growth & sediment build-up

Once a dummy is planted, a green bar above the clam fills up. Growth time depends on your level: 2 seconds at Level 1, 10 at Levels 2 and 3, and 30 from Level 4 on (Faster Growth upgrades bring it down, but never below 5 seconds). From Level 2 on, sediment builds up on growing clams and slows them down, up to 90% slower when fully dirty. It never stops growth completely though. You can see it as a grimy overlay that gets darker the dirtier the clam is.

Scrubbing (Brush)

Press 2 or B and click a dirty clam. Each click takes off half of the max sediment, so two clicks cleans a fully dirty clam. When a clam is scrubbed all the way clean (or just planted) it gets a 4 second break before dirt starts building again. Scrubbing gives no XP on purpose, it's basically free growth speed.

Chum & the chum meter

Press 3 or C and click a growing clam to instantly add 3 seconds of growth (plus 2 XP). It only works on growing clams, not empty or ready ones. The catch is that every click adds 0.5 to the chum meter in the top right (it goes from 0 to 10 and slowly drains at 0.08 per second, so a full meter takes about 2 minutes to empty). Once the meter reaches the white tick, crabs start coming, which takes about 8 to 10 chum clicks.

Crabs (targeting, attack, pearl loss)

Crabs unlock at Level 4. When the chum meter is at or above the tick, crabs walk in from the left or right side of the screen, and they spawn faster the fuller the meter is. Each crab picks a random clam that has a pearl in it, either still growing or finished and waiting to be harvested, so crabs usually spread out across the pond. A crab only picks a new target if its clam ends up empty. If a crab

reaches its clam, growth freezes and the bar turns red. If the red bar fills (2 seconds), the pearl and any sediment are gone, the clam is empty again, and the crab runs off. Harvesting a finished pearl while it's being attacked saves it and sends the crab off to a new target.

Sonic flares (radius, ammo, buying more)

Press 4 or F and click anywhere in the water to fire a sonic flare. Every crab inside the radius runs away right away, and if one was attacking, a growing pearl goes back to growing from where it was and a finished pearl stays ready to harvest. The shockwave animation is scaled to the real radius (250 px, up to 500 px with upgrades), so you can see exactly what you hit. You get 5 sonic flares at Level 4, 1 more every level after that, and you can buy more for \$100 each in the shop. Your ammo count is on the sonic flare slot of the toolbar.

Harvesting & selling at the surface dock

When a clam opens up and shows a pearl, click it with the Hand to harvest it (3 XP). It goes into your pearl counter in the bottom right. Don't leave it sitting there too long, since crabs go after finished pearls too. Then click the surface dock (the one with the [SELL HERE] sign) with the Hand to sell everything you're carrying. Each pearl type has its own price and XP, from \$30 and 40 XP for a Classic up to \$230 and 180 XP for an Abalone.

XP, leveling, and unlocks

Your level and XP bar are in the top left. In Stage 1, each level needs a certain number of pearls sold (1, 2, 3, then 3), so you can't level up just by clicking around. In freeplay, each level costs a bit more XP than the last. Every level gives you something: a new tool in Stage 1, then an upgrade pick every freeplay level, a new pearl type on odd levels from 5 to 15, and a free sonic flare every level after 4.

Pearl types & rarity

There are 7 pearl types based on real ones: Classic (Akoya), Blush, Tahitian, Golden South Sea, Conch, Melo, and Abalone. When a pearl finishes growing, the game rolls which type it is out of the ones you've unlocked, and rarer ones are less likely (with everything unlocked it's 40% Classic down to 2% Abalone). The growing dummy always looks the same, so you don't know what you got until the clam opens. By Level 15, every pearl type is unlocked.

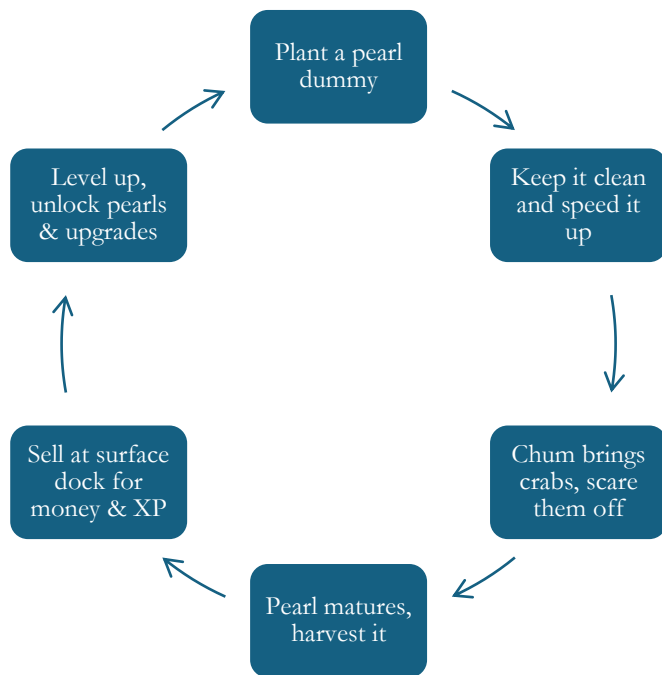
Upgrades & shop

There are 4 upgrades: New Clam Slot, Faster Growth, Bigger Sonic Flare, and Crab Tolerance (raises the chum meter line where crabs start spawning). Every freeplay level you get to pick 1 of 3 random upgrades, then either Buy Now or Save for Later. The shop button (top right) has three tabs: Available (saved picks), Owned (what you've bought and the total effect), and Supplies (sonic flares). Each upgrade gets pricier every time you buy it, and each one has a cap so the game doesn't get too easy.

Saving / loading

Hit Esc or the pause button, then Save or Load. There's one save slot. It saves your level, XP, money, sonic flares, pearls you're carrying, upgrades, and every clam's state. Crabs, the chum meter, and the settings toggles aren't saved and reset when you load.

5.3 Core Gameplay Loop



The whole game is basically this loop over and over, but it gets harder to keep up with as you get more clams and more crabs at the same time.

5.4 Step-by-Step Walkthrough for a New Player

- Step 1: A welcome popup explains the basics. Close it and you're at Level 1 with one clam and the Hand selected. Click the clam to plant a pearl dummy.
- Step 2: Wait about 2 seconds for the green bar to fill. When the clam opens, click it to harvest the pearl.
- Step 3: Click the surface dock at the top (the [SELL HERE] sign) to sell it. That gets you to Level 2.
- Step 4: Level 2 unlocks the Brush and a second clam. Plant both, and when they get dirty, press 2 or B and scrub them. Sell 2 pearls to level up.
- Step 5: Level 3 unlocks Chum and a third clam. Growth is still 10 seconds, but press 3 or C and chum your growing clams to speed them up even more. Keep an eye on the chum meter. Sell 3 pearls.
- Step 6: Level 4 unlocks crabs and Sonic Flares (you get 5), and growth goes up to 30 seconds. If you chum past the tick on the meter, crabs show up and go after any clam with a pearl, even finished ones. Press 4 or F and click near them to scare them off, and harvest finished pearls fast. The shop button also shows up here. Sell 3 more pearls.
- Step 7: Level 5 starts freeplay. You unlock the Blush pearl and get your first upgrade pick. Choose a card, then Buy Now or Save for Later.
- Step 8: From here, keep the loop going: balance chum against crabs, buy upgrades, unlock rarer pearls, and work your way up to max level (Level 30). You can save from the pause menu whenever.

6. Rules

6.1 Objective / Win Condition

There are really two goals. The first one is finishing Stage 1, which means getting to Level 5 by selling pearls (1, 2, 3, and 3 pearls for each level-up).

After that it's freeplay, and the "win" is hitting max level. That happens once every pearl type is unlocked and every upgrade is maxed out (bought or saved up to its cap), which with the current numbers is Level 30. The XP bar fills up and says MAX LEVEL. There's no game over or end screen, so you can keep farming after that if you want.

6.2 Lose / Setback Conditions

You can't really lose *Clam It!*, but you can have setbacks:

- If a crab finishes its attack, the pearl is destroyed, all its growth progress is gone, any sediment is cleared, and the clam goes back to empty. This can happen to finished pearls you haven't harvested yet, too. You have to plant a new dummy and start over on that clam.
- You can run out of sonic flares. If you're also out of money, the only option is to stop chumming and let the meter drain back below the line.
- Money can't go negative. Buy buttons are just disabled when you can't afford something.
- Letting clams get dirty doesn't kill them, it just makes them grow really slowly (10% speed when fully dirty).

6.3 Resource Rules

- **Money:** you only earn it by selling pearls (price depends on the type). You spend it on upgrades and sonic flares (\$100 each). Upgrade costs go up each time you buy the same one (cost = base cost $\times$ multiplier to the power of how many you own).
- **XP:** plant = 2, harvest = 3, chum = 2, scrub = 0, sell = 40 to 180 per pearl depending on type. Stage 1 needs 45, 135, 270, and 405 total XP for Levels 2, 3, 4, and 5. In freeplay, each level costs $300 + 60 \times N^{1.5}$ XP ($N = 1$ for the Level 5 to 6 step), so 360, 469, 611, and so on.
- **Sonic flares:** you start with 0, get 5 at Level 4, then 1 more every level-up after that. You can buy more for \$100 each in the Supplies tab of the shop. The Infinite Sonic Flares setting removes the limit.
- **Clam slots:** 1 clam at Level 1, 2 at Level 2, 3 at Level 3. The New Clam Slot upgrade adds up to 3 more (\$600, then \$1200, then \$2400), for a max of 6. Each clam has a fixed spot on the sea floor.

6.4 Clam & Growth Rules

Every clam is always empty, growing, or ready. Here's how they work:

- Growth time is 2 seconds at Level 1, 10 at Levels 2 and 3, and 30 from Level 4 on. Each Faster Growth upgrade takes off 2 seconds, but it never goes below 5 seconds.
- Growth only happens while a clam is growing. Growth speed = 1 minus $(0.9 \times \text{sediment})$, so a fully dirty clam still grows at 10% speed.

- Sediment only builds up from Level 2 on, only on growing clams, at 0.15 per second, up to 1.0. Right after planting or scrubbing a clam fully clean, there's a 4 second grace period with no build-up.
- The Brush removes 0.5 sediment per click. Using it on a clean clam just says "Clam is already clean!"
- Chum adds 3 seconds of growth per click and only works on growing clams. If that finishes the pearl, it matures right away.
- The Hand plants on empty clams, harvests on ready clams, and does nothing on growing ones.
- The pearl's type is picked the moment it matures, so the open clam already shows the right pearl.

6.5 Crab Rules

Crabs are the main threat in the game, and they follow these rules:

- Crabs can only show up from Level 4 on.
- The chum meter goes from 0 to 10. Each chum click adds 0.5, and it drains by 0.08 per second no matter what. That means about 8 to 10 chum clicks before crabs start showing up.
- Crabs spawn when the meter is at or above the line (4 by default, up to 7 with Crab Tolerance upgrades), there's room for another crab, and at least one clam has a pearl in it (growing, or finished and not harvested yet).
- The fuller the meter, the faster they spawn: every 6 seconds right at the line, down to every 1.5 seconds when it's full.
- They come in from the left or right edge at a random height in the water.
- Max crabs on screen starts at 4 and goes up by 1 every 3 levels after Level 5 (so 5 at Level 8, 6 at Level 11, and so on).
- Each crab walks at 100 px/s toward a random clam that has a pearl in it, growing or finished. A pearl finishing doesn't change the crab's target. It only picks a new random target when its clam becomes empty (harvested, or destroyed by another crab). If no clam has a pearl, it leaves.
- When a crab reaches a clam, growth and sediment freeze and a red bar fills over 2 seconds. When it's full, the pearl is destroyed and the crab runs off screen at double speed. Finished pearls can be attacked too, but if you harvest one mid-attack, you keep the pearl and the crab goes after a new target.
- A sonic flare scares off every crab inside its radius (250 px, +50 per Bigger Sonic Flare upgrade, max 500 px). An attacking crab lets go: a growing pearl keeps growing from where it was, and a finished pearl stays ready to harvest.

6.6 Leveling, Pearl & Upgrade Rules

These are the rules for leveling, pearls, and upgrades:

- **Tool unlocks:** Hand at Level 1, Brush at 2, Chum at 3, Sonic Flare (and crabs) at 4. Trying to use a locked tool just posts a warning in the chat log.
- **Pearl unlocks:** Classic from the start, then Blush at 5, Tahitian at 7, Golden South Sea at 9, Conch at 11, Melo at 13, Abalone at 15. After 15, odd levels only give the upgrade pick. So by Level 15, all 7 pearl types are unlocked. If you want to see them all without playing that far, load the all-unlocked save from section 3.3.

- **Pearl prices:** Classic \$30, Blush \$45, Tahitian \$65, Golden South Sea \$95, Conch \$130, Melo \$175, Abalone \$230. Rarer pearls also give more XP when sold.
- **Drop rates:** only unlocked types can show up, and each one's chance is its weight divided by the total weight of what's unlocked. With everything unlocked: Classic 40%, Blush 22%, Tahitian 15%, Golden South Sea 10%, Conch 7%, Melo 4%, Abalone 2%.
- **Upgrade picks:** every freeplay level shows 3 different random upgrades that aren't maxed out yet. You pick one, then Buy Now (if you can afford it) or Save for Later (goes to the shop's Available tab). Saved picks count toward the cap, and if fewer than 3 types are left you see fewer cards.
- **Upgrade caps and prices:** New Clam Slot, 3 max, starts at \$600 and doubles each time. Faster Growth, 12 max, starts at \$300 ($\times 1.5$ each time). Bigger Sonic Flare, 5 max, starts at \$250 ($\times 1.7$ each time). Crab Tolerance, 6 max, starts at \$250 ($\times 1.6$ each time). Every upgrade is permanent. Early in freeplay, one level's worth of pearls pays for about one cheap upgrade, so the last few stacks of each upgrade are long-term goals.
- **Popups:** on a level that unlocks a pearl, the pearl popup comes first and then the upgrade pick. If you level up more than once from one sale, all the popups line up and the game stays paused until you close the last one.
- **Max level:** once all pearls are unlocked and all upgrades are maxed, leveling stops (Level 30 right now).

7. Level Design & Progression

7.1 Level Overview

Level / Stage	Purpose	Abilities introduced	Est. playtime
Stage 1: Tutorial (Lvl 1 to 4)	Teach every tool one at a time, with nothing too stressful going on	Hand (planting, harvesting), selling at the dock, Brush, Chum and the chum meter, Sonic Flares and crabs	About 3 to 5 min
Stage 2: Freeplay (Lvl 5 to 30)	Use all the tools together with more clams, more crabs, and money decisions	Rare pearl types, upgrade picks, the shop's Available and Owned tabs	First few levels take about 1 to 2 min each; max level is roughly an hour

7.2 Stage 1: Learning the Ropes

Stage 1 is a guided tutorial where each level adds exactly one new thing, and you can only level up by selling pearls:

- **Level 1:** one clam, just the Hand. Growth takes only 2 seconds. You plant, harvest, and sell one pearl, which teaches the basic loop.
- **Level 2:** the Brush unlocks, a second clam opens up, and sediment starts building. Growth is 10 seconds. Now you have to keep two clams clean at once. Sell 2 pearls.
- **Level 3:** Chum unlocks, a third clam opens up, growth is still 10 seconds, but chum becomes the way to speed things up. The chum meter starts filling, but it's safe for now. Sell 3 pearls.
- **Level 4:** crabs and Sonic Flares unlock, you get 5 flares, and the shop button appears. Growth reaches its intended speed of 30 seconds. Now everything you learned has to work together, since chumming too much brings crabs. Sell 3 more pearls to finish Stage 1.

7.3 Stage 2: Applying the Skills

Freeplay doesn't teach any new tools. Instead, it makes you use the ones you already know in new ways:

- **More clams at once:** with up to 6 clams, you can't babysit each one. You have to decide which clams get scrubbed and chummed first.
- **Chum becomes a real gamble:** crabs spawn faster the fuller the meter is, and the max number of crabs goes up every 3 levels. Chumming is now about managing the meter, not just clicking for speed.
- **Sonic flares turn into aiming and saving:** with more crabs on screen and limited ammo, it's about hitting as many crabs as you can with one sonic flare, and choosing between buying sonic flares and saving for upgrades.
- **Protecting the right pearl:** rare pearls are worth a lot more, and since crabs pick random clams (finished ones included), a rare pearl left sitting in an open clam is a big risk. Some pearls are way more worth defending than others, and harvesting fast matters.
- **Upgrades change your strategy:** Crab Tolerance lets you chum more, Bigger Sonic Flare makes defense easier, Faster Growth and extra clams speed everything up. What you pick changes how you play.

7.5 Difficulty Curve & Playtime

The difficulty goes up slowly. Stage 1 adds one tool per level, and the growth time goes from 2 to 10 to 30 seconds, so each level takes a bit longer and needs more attention. Crabs only show up after you know every other tool. In freeplay, each level costs more XP than the last, and every 3 levels one more crab can be on screen, while upgrades give you ways to push back.

8. Art, Animation & Audio

8.1 Visual Style

I really like games with hand-drawn, janky art styles, so I wanted to use that kind of look for a simple game like this. Drawing the assets by hand was also free and without any possible copyright issues. For font, I used a fun font I found on Google Fonts a while ago, Slackey, which I am pretty sure is the SpongeBob font (or at least it's very similar), so it matched the underwater theme perfectly.

8.2 Animations

The game has only 2 animations: crab walking and sonic flare. The latter is an important part of the gameplay, as it has to visually indicate how big the flare's radius is. It's a simple 5-frame expanding animation, and I made it fade away (tween) with flare_effect.gd. Same tween function was also used for floating text when selling pearls at the dock.

8.3 Audio

Event	Sound
Collecting a pearl	collect.mp3
Crab attack / pearl destroyed	destroy.mp3
UI elements, switching tools	select.mp3
Crab walking	crab_walk.mp3

All audio tracks were recorded and produced by me.

9. Design Schemas (Rules of Play)

Schema	Definition	How <i>Clam It!</i> uses it
Games as Emergent Systems	A few simple rules that, once they interact with each other (and with what the player decides to do), create gameplay you couldn't predict just by reading the rules. The whole is more than the sum of its parts.	Every rule put together create situations I never programmed directly. Crabs end up spread all over the pond, so you're always figuring out where one sonic flare can cover the most clams. Leaving a finished pearl out while you go chum another clam can get it destroyed, and harvesting a pearl mid-attack saves it but sends that crab off to a new random clam, maybe the one you just chummed. With 6 clams and more crabs later on, it turns into a juggling act that plays out differently every time.
Games as Systems of Uncertainty	A game needs some uncertainty about what's going to happen, or there's nothing to learn or decide. It doesn't have to be pure randomness. "Risk" is when you know the odds ahead of time, and the goal is to make things feel random while skill still matters.	Pearl types are rolled with odds you can know ahead of time (40% to 2%), which is risk in the book's sense, and since the growing dummy always looks the same, you don't know what you got until the clam opens. Crabs come in from a random side at a random height, each one picks a random clam to go after, and the upgrade pick shows 3 random choices. But skill still decides how things go: how much you chum, how fast you harvest, where you aim sonic flares, and what you buy.
Games as Cybernetic Systems	A system with a sensor, a comparator, and an activator: the game watches its own state, decides if something needs to change, and changes it. Positive feedback loops reward you, while negative feedback loops push back and keep things balanced.	The chum meter is a textbook example. The sensor is the meter value, the comparator is the white line, and the activator is crab spawning, which gets faster the fuller the meter is. That's a negative feedback loop. The more you speed things up, the more the game pushes back. Sediment is another one. The positive loop is selling pearls for money and XP, buying upgrades like more clams and faster growth, and making even more pearls. I dampen that with upgrade caps, rising prices, a rising XP curve, and more crabs every 3 levels.
Games as Systems of Conflict	Conflict is about who you're up against. It can be player vs. player, group vs. group, one vs. many, everyone for themselves, or a single player against the game system.	<i>Clam It!</i> is a single player against the game system. Crabs are the direct conflict: they go after your pearls and you fight back with flares. There's also indirect conflict with the environment (sediment slowing you down) and conflict over limited resources, since flares and money are limited and you have to decide where they go. There's no multiplayer or high score, so the closest thing to a score is your level.

10. Iterations & Playtesting

10.1 Playtesting Process

I sent .exe builds to a few of my friends that weren't in this class, and watched some play it, while others just sent me a list of things to add. When I watched my friends play it, I noted down a lot of changes I needed to make myself, without them having to tell me.

10.2 Iteration Log

Commit	What changed	Why (playtest feedback / reasoning)
49890a9	Made crabs target random pearls and matured pearls; fixed a bug where a pearl remained under attack if crab was scared off	Multiple crabs were targeting the same pearl, and having them flee once a pearl matured made the game too easy
fe784bc	Created an intro popup that explains how the game is played; gave the upgrades menu more spacing and padding so it looks better; increased the size of various popups and menus so it's easier to read	Playtester reported UI was really janky and too small, and that I should add an intro popup
35e7998	I shortened Stage 1 clam growth times so the player gets to Stage 2 sooner	Playtester said Stage 1 takes too long and he got bored before he even discovered Stage 2
0700127	Added [SELL HERE] to the dock to indicate that you sell there	Playtester suggestion. I didn't have enough time and energy to implement what I actually wanted there so this placeholder will do
b57e275	Increased crab speed; reduced money for pearls and increased upgrade multiplier cost; balanced chum meter	My own playtesting led me to realize that the game was super unbalanced when it came to money and difficulty. I played around with a bunch of values to find some that make the game at least a little more challenging

10.3 Key Design Changes

Besides adding features, a lot of the iterations were tuning numbers after playing and noticing things felt off. The biggest ones:

No more XP for scrubbing. Scrubbing used to give XP, but it made players level up way too early just by clicking dirty clams. I removed it and made the Stage 1 level thresholds based on how many pearls you sell, so the only way to level up is actually selling pearls.

Chum became an instant jump. Originally chum was planned as a temporary speed boost. I changed it to an instant jump in growth, since it's way easier to see and feel. It started at 1 second per click, which barely did anything, so I bumped it to 3 seconds.

Scrubbing got stronger and less annoying. Each scrub used to take off 0.2 sediment, so cleaning a clam took forever. Now it's 0.5 per click, and there's a 4 second grace period after planting or cleaning so you're not scrubbing nonstop.

Crab tuning. The crab threshold went from 3 to 4 on the chum meter so a little chum is safe. Each chum click used to add 2 to the meter, which meant crabs showed up after just a couple clicks, so I dropped it to 0.5 per click and slowed the drain down to 0.08 per second. Now it takes about 8 to 10 clicks to bring crabs in, but the meter stays high for a while once you do. Crabs spawn faster as the meter fills, and in freeplay the max number of crabs goes up every 3 levels so crabs stay a threat once you stack defensive upgrades. The sonic flare also changed from something you click on a clam to something you fire anywhere in the water.

Economy rebalance. Pearls used to sell for way too much (a Classic was \$200), so money piled up and you could buy upgrades as soon as you picked them. I cut pearl prices way down (Classic is now \$30, Abalone \$230), raised sonic flares from \$30 to \$100, and made upgrade prices climb faster with each purchase. Now early upgrades take a level or so of saving, and the last stacks of each upgrade are real long-term goals.

11. Builds & Installation

Platform	File on release	How to run
Windows (x86)	ClamIt_Windows.zip (contains ClamIt.exe)	Unzip it and double-click ClamIt.exe. If Windows SmartScreen pops up, click More info, then Run anyway (the game isn't signed).
Mac	ClamIt_Mac.zip (contains ClamIt.app)	Unzip it, then right-click ClamIt.app and choose Open the first time (macOS blocks unsigned apps on a normal double-click).
Linux (x86)	ClamIt_Linux.zip (contains ClamIt.x86_64)	Unzip it, make the file executable (chmod +x ClamIt.x86_64, or tick "Allow executing" in file properties), then run it.

12. Credits & Assets

All sprites and audio were created by me, Dani Zhumasheva.

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13. Future Work

There's a bunch of stuff I'd like to add if I keep working on this:

- **Marketplace at the dock:** spawn customers that want different amounts and combinations of pearls, so you have to wait to have certain pearls before selling them
- **Free clam placement:** let players put new clam slots anywhere in the pond instead of in fixed spots.
- **More upgrade types:** the pick and shop systems already read from a data table, so new ones would be easy to plug in.
- **Flawless and Radiant pearls:** recolored versions of existing pearls (palette swap plus sparkles) so you keep getting a new pearl every other level after Level 15.
- **Pearl collection log:** a stats screen showing every pearl type you've ever harvested.
- **Pond decorations:** cosmetic stuff to buy, so there's something else to spend money on besides upgrades.
- **Idle and offline growth:** clams keep growing even when the game is closed.
- **Audio settings:** volume controls in the settings menu.